

DBFuzz Implementation on Doctor Appointment Management System (DAMS)

Sandeep Khachariya ID: 2023UCP1628
Dhiraj Chouhan ID: 2023UCP1591

MNIT Jaipur Project Presentation

Project Title: DBFuzz Implementation for Stored XSS Testing

Objective:

- Deploy DBFuzz research prototype
- Integrate with DAMS web application
- Perform automated login, crawling, and fuzz testing
- Detect potential stored XSS vulnerabilities

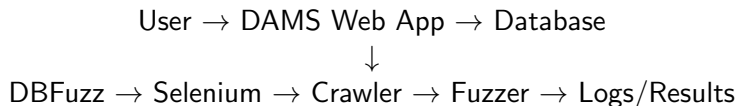
Repository Used: <https://github.com/Spider-Scents/dbfuzz>

Problem Statement

Manual security testing of web applications is time consuming.

DBFuzz automates:

- Authentication handling
- Session cookie extraction
- Crawling hidden authenticated pages
- Payload injection
- Vulnerability detection



Technologies Used

- Ubuntu Linux
- Python 3.10
- Pipenv
- Docker + Docker Compose
- MariaDB
- phpMyAdmin
- Chromium + ChromeDriver
- Graphviz
- DBFuzz Framework

Implementation Steps

- 1 Clone DBFuzz repository
- 2 Install dependencies
- 3 Setup Docker environment
- 4 Deploy DAMS application
- 5 Import SQL database
- 6 Configure DBFuzz
- 7 Run automated scanning

Docker Deployment

Commands used:

Start Containers

```
cd /Desktop/dbfuzz/docker/dams  
sudo docker-compose up -d
```

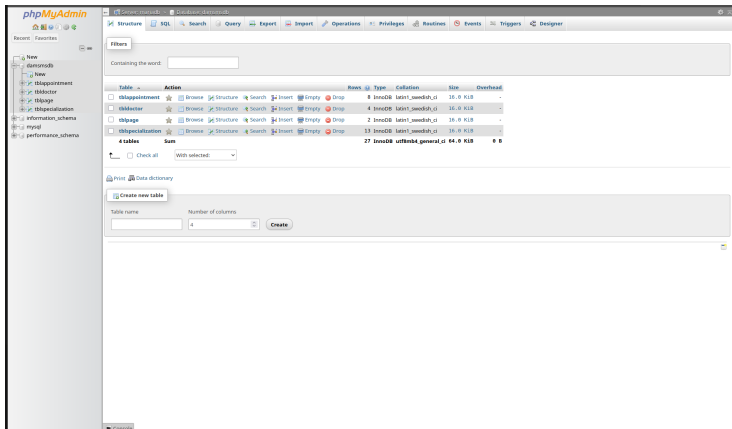
Containers started:

- Web Server
- MariaDB
- phpMyAdmin

```
starlink@Ubuntu:~/Desktop/dbfuzz/docker/dams$ sudo docker-compose up -d  
[sudo] password for starlink:  
dams_phpmyadmin_1 is up-to-date  
dams_mariadb_1 is up-to-date  
dams_web_1 is up-to-date  
starlink@Ubuntu:~/Desktop/dbfuzz/docker/dams$
```

Database Setup

- Open phpMyAdmin
- Create database: damsmsdb
- Import damsmsdb.sql



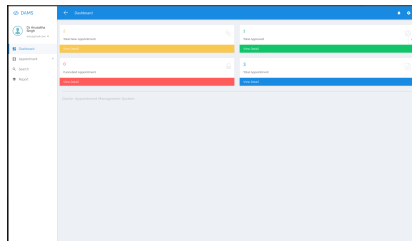
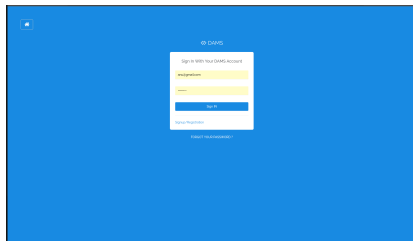
DAMS Web Application

URLs:

- <http://127.0.0.1/dams>
- <http://127.0.0.1/dams/doctor/login.php>

Credentials:

- Email: anu@gmail.com
- Password: Test@123



DBFuzz Configuration

Command:

Run Scanner

```
pipenv run python dbfuzz.py --config docker/dams/config_doctor.ini
```

Configuration includes:

- Target URL
- Login URL
- Credentials
- Database details

Challenges Faced

- Docker Compose variable issues
- Outdated Debian repositories
- Broken Dockerfile
- Disk storage full
- Cookie extension failure
- Chrome compatibility issues
- Browser automation failures

Novel Work / Modifications

- Fixed Docker Compose variables
- Updated outdated repositories
- Rebuilt Dockerfile
- Fixed storage issues
- Replaced broken cookie extension logic
- Modified Browser.py
- Direct Selenium cookie extraction
- Fixed Chromium compatibility

Modified Browser.py Logic

Original issue:

- Cookie extension unsupported

Solution:

- Directly extract cookies using Selenium

Scanning Process

DBFuzz performed:

- Automated login
- Session extraction
- URL crawling
- Endpoint discovery
- Fuzz testing

```
assert len(files) > 0
AssertionError
starlink@ubuntu: ~/Desktop/dbfuzz$ nano ~/Desktop/dbfuzz/Browser.py
starlink@ubuntu: ~/Desktop/dbfuzz$ nano ~/Desktop/dbfuzz/Browser.py
starlink@ubuntu: ~/Desktop/dbfuzz$ cd ~/Desktop/dbfuzz
pipenv run python dbfuzz.py --config docker/dans/config_doctor.ini
Fuzzing started at 2026-05-09 15:17:51.463282
BACKWARDS COMPATIBILITY: adding match=0, to credentials
login - url http://localhost/dans/doctor/login.php - credentials match=0,email=amu@gmail.com,password=Test@123
{"download.default_directory": "/home/starlink/Desktop/dbfuzz/cookies"}
Waiting 20 seconds for cookie download
login URL: http://localhost/dans/doctor/cancelled-appointment.php
cookies, user agent, and login URL: {"cookie_str": "PHPSESSID=095af7707e3384e38ecb28f344c88daf", "user_agent": "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36", "url": "http://localhost/dans/doctor/cancelled-appointment.php"}
Scanning for list of URLs to check
Initial Scanning http://localhost/dans/doctor/dashboard.php
python3 -u /home/starlink/Desktop/dbfuzz/blackwidow/crawl.py --shallow http://localhost/dans/doctor/dashboard.php --cookies PHPSESSID=095af7707e3384e38ecb28f344c88daf
python3 -u /home/starlink/Desktop/dbfuzz/blackwidow/crawl.py --shallow http://localhost/dans/doctor/dashboard.php --cookies PHPSESSID=095af7707e3384e38ecb28f344c88daf
BM: b'adding cookie PHPSESSID = 095af7707e3384e38ecb28f344c88daf\n'
BM: b'-----\n'
BM: b'Edges left: 1\n'
BM: b'\n'
BM: b'-----\n'
BM: b'GETS | FROMS | EVENTS\n'
BM: b'1 | 0 | 0\n'
BM: b'-----\n'
BM: b'-----\n'
BM: b'Crawl (edge): [get] ROOTREQ -(get None[False])> [get] http://localhost/dans/doctor/dashboard.php\n'
BM: b'-----\n'
BM: b'Edges left: 24\n'
BM: b'\n'
BM: b'-----\n'
BM: b'GETS | FROMS | EVENTS\n'
BM: b'14 | 0 | 9\n'
BM: b'-----\n'
BM: b'-----\n'
BM: b'Crawl (edge): [get] http://localhost/dans/doctor/dashboard.php -(get None[False])> [get] http://localhost/dans/doctor/search.php\n'
BM: b'-----\n'
BM: b'Edges left: 48\n'
BM: b'\n'
BM: b'-----\n'
BM: b'GETS | FROMS | EVENTS\n'
BM: b'27 | 0 | 19\n'
BM: b'-----\n'
BM: b'-----\n'
BM: b'Crawl (edge): [get] http://localhost/dans/doctor/dashboard.php -(get None[False])> [get] http://localhost/dans/doctor/profile.php\n'
BM: b'-----\n'
```

Results

Generated Output:

- Crawl logs
- Endpoint discovery logs
- Error logs

Important file:

blackwidow/logs/crawl-1778320121.7225757.log

[illegible][illegible]

Detected Endpoints

Examples:

- dashboard.php
- profile.php
- search.php
- appointments.php

These were automatically discovered by crawler.

Conclusion

- Successfully deployed DBFuzz
- Integrated with DAMS
- Automated login + crawling completed
- Generated scan logs
- Improved original outdated implementation

Thank You

Questions?